

Chapter 4

Evidence-Aware Evaluation of Question Decomposition

Introduction

Chapter 3 examined how compositional difficulty is represented before a model answers a question. This chapter studies the retrieval process that follows a decision to decompose. Its objective is to measure whether the resulting subquestions retrieve the evidence needed for the original question, and how deeply each relevant retrieval list must be searched.

A decomposition changes several parts of the computation at once. It creates additional retrieval queries, assigns information needs to those queries and passes intermediate answers between them. Final-answer accuracy combines these effects. Passage recall measures evidence availability but leaves the assignment of that evidence to individual subquestions unspecified. Evaluating the decomposition therefore requires three separate quantities: complete evidence availability, evidence ownership and successful answer execution.

The chapter develops an evaluation based on a matrix of evidence ranks and a learned ownership mask. The mask identifies which supporting passages each subquestion is intended to retrieve. A maximum over the selected cells measures the depth required to satisfy those assignments. A separate union measurement records whether the evidence was retrieved anywhere in the execution. The ownership model is evaluated on reference-style decompositions whose assignments are known. Retrieval and answer results are then compared on a 100-question MuSiQue panel. Finally, fixed-list budget replay distinguishes per-query depth from total passage expenditure. These measurements provide the objective and diagnostic information used for prompt optimisation in Chapter 5.

4.1 Problem Formulation

4.1.1 A Decomposition as an Executable Sequence

Let q be a question with reference answer y and annotated supporting passages $E(q) = \{e_1, \dots, e_h\}$. MuSiQue supplies both supporting passages and a reference decomposition into connected single-hop questions [1]. A generated decomposition is an ordered sequence $U(q) = (u_1, \dots, u_A)$. An item u_a may contain a reference to the answer of an earlier item. The execution substitutes the predicted earlier answer, retrieves passages for the resulting

query $\tilde{u}_a$ and produces an intermediate answer $\hat{y}_a$:

$$\tilde{u}_a = \text{substitute}(u_a, \hat{y}_1, \dots, \hat{y}_{a-1}), \quad \hat{y}_a = \text{reader}(\tilde{u}_a, R_a[1 : k]). \quad (4.1)$$

Here $R_a = (d_{a1}, d_{a2}, \dots)$ is the ranked passage list returned for $\tilde{u}_a$, and k is the number of passages supplied to the reader. Backward references make the execution sequential even when the decomposition contains conceptually independent branches. The retained implementation returns $\hat{y}_A$ as the final answer. Each step receives its own retrieved context; the final step has no additional synthesis call over the union of all earlier passages. One such step, a resolved query with its retrieval and its reader call, is an arm of the execution.

The number of generated subquestions A may differ from the reference hop count h . One subquestion can combine several reference information needs. Several subquestions can address the same need. The evaluation therefore permits many-to-many assignments instead of requiring an alignment by position.

4.1.2 Availability, Assignment and Execution

The three evaluation objects are defined as follows.

Availability. Every annotated supporting passage occurs in at least one retrieved prefix.

Assignment. Every supporting passage has an identified owner, and each owning subquestion retrieves the passages assigned to it.

Execution. The reader produces the required intermediate answers and the final answer matches the reference under the specified answer metric.

These properties answer different questions. Evidence retrieved for a later subquestion may have been unavailable when an earlier answer was produced. Conversely, a model may answer correctly from a passage outside the annotated evidence set or from parametric knowledge. The measurements below concern recovery of benchmark evidence and reference-answer accuracy. Their relation is examined empirically rather than imposed by definition.

4.2 Retrieval Depth and Evidence Responsibility

4.2.1 The Evidence-Rank Matrix

For a recorded execution, define an $A \times h$ matrix

$$\rho_{ai} = \min\{r : d_{ar} = e_i\}. \quad (4.2)$$

The evaluation retains the first $K = 200$ ranked passages per subquestion. When e_i is absent from that prefix, its recorded rank is $+\infty$. This value denotes censoring at K , including evidence that might occur deeper in the full retrieval list. Passage matching uses the benchmark passage identity, determined from its title and text.

The matrix records where each required passage occurs for each query. It does not determine which query was responsible for that passage. That second object is introduced below.

4.2.2 Union Depth

The union depth is the smallest common prefix length that makes all annotated evidence available somewhere in the retrieval lists:

$$D_{\text{union}}(q) = \max_{1 \leq i \leq h} \min_{1 \leq a \leq A} \rho_{ai}. \quad (4.3)$$

Consequently, $D_{\text{union}}(q) \leq k$ exactly when $E(q) \subseteq \bigcup_a R_a[1 : k]$. The inner minimum allows one successful retriever to satisfy the availability requirement for a passage. The outer maximum requires every annotated passage to be present.

This definition is appropriate for measuring a pooled evidence collection. In the sequential execution of Equation 4.1, however, a passage can appear in the pooled collection without reaching the particular reader call that needed it. Union depth is therefore reported alongside an assignment-aware measurement.

4.2.3 Ownership-Masked Depth

Let $M \in \{0, 1\}^{A \times h}$ be an ownership mask, with $M_{ai} = 1$ when subquestion a is assigned responsibility for passage e_i . Define

$$D_M(q) = \begin{cases} \max_{(a,i): M_{ai}=1} \rho_{ai}, & \text{if every column of } M \text{ contains an owner,} \\ +\infty, & \text{otherwise.} \end{cases} \quad (4.4)$$

An assigned cell whose evidence is absent from the retained list also makes $D_M(q) = +\infty$. The evaluation distinguishes this retrieval censoring from an unowned column, which indicates an uncovered assignment under the predicted mask.

The row and column structure has a direct interpretation. A zero row is an idle subquestion with respect to the annotated evidence. A row with several ones assigns several evidence items to one subquestion. A zero column leaves an evidence item unowned. A column with several ones assigns that evidence to several subquestions. Idle rows contribute no retrieval obligation to Equation 4.4; their generation and reading costs remain part of the executed system.

Multiple ownership is interpreted conjunctively: each selected owner must retrieve the passage. A rule that accepts retrieval by any selected owner would use a minimum over owners and would measure a different execution requirement. The present rule preserves the interpretation of each positive cell as a specific obligation of a specific reader call.

For comparison, the all-cells maximum is

$$D_{\text{all}}(q) = \max_{a,i} \rho_{ai}. \quad (4.5)$$

It requires every subquestion to retrieve every supporting passage. Whenever the mask assigns every evidence item, the three quantities satisfy

$$D_{\text{union}}(q) \leq D_M(q) \leq D_{\text{all}}(q). \quad (4.6)$$

The first inequality follows because each selected rank is at least the smallest rank in its column. The second follows because the masked maximum ranges over a subset of all cells. A zero column is handled separately by the censoring rule in Equation 4.4.

For example, consider

$$\rho = \begin{pmatrix} 1 & 80 \\ 40 & 2 \end{pmatrix}, \quad M = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}. \quad (4.7)$$

Both union depth and ownership-masked depth equal 2, while the all-cells maximum equals 80. Each subquestion retrieves its own assigned passage near the beginning of its list. Requiring the off-diagonal passages changes the question being evaluated. With a single full-question query, the control assigns all evidence to that query, so the three depth definitions coincide.

4.2.4 Coverage and an Aggregate Depth Score

For a panel of n questions, complete-evidence coverage at depth k is

$$C_D(k) = \frac{1}{n} \sum_{j=1}^n \mathbf{1}\{D(q_j) \leq k\}, \quad (4.8)$$

where D is either D_{union} or D_M . Coverage is a question-level complete-set measurement. A question missing one required passage receives the same coverage value as a question missing several, while the retained rank matrix identifies the individual missing items.

Prompt optimisation also requires a scalar objective. The retained objective assigns logarithmic credit to finite depths:

$$v_K(d) = \begin{cases} 1 - \frac{\log d}{\log K}, & 1 \leq d \leq K, \\ 0, & d = +\infty, \end{cases} \quad J_D = \frac{100}{n} \sum_{j=1}^n v_K(D(q_j)). \quad (4.9)$$

The score is 100 when all required evidence appears at rank 1 and 0 when every question is censored or reaches depth K . Questions with unowned evidence remain in the denominator. Reporting $C_D(K)$ beside J_D distinguishes incomplete retrieval from complete retrieval at the search limit. The score summarises complete coverage and depth. Arm count and model tokens are recorded separately because Equation 4.9 does not price them.

4.3 Learning the Ownership Mask

4.3.1 Supervision from Reference Decompositions

MuSiQue associates each reference subquestion with a supporting passage. Substituting the reference intermediate answers resolves its dependencies and provides labelled subquestion–passage pairs. The associated passage is a positive ownership example; the other supporting passages of the same original question provide negative examples. These labels encode the benchmark’s reference assignment. An alternative valid decomposition can have a different assignment.

Training uses a curated set of 6 847 MuSiQue training questions. The resulting training matrix contains 36 694 cells, of which 15 452 are positive. The held-out control contains 100 questions with reference-style subquestions and known reference assignments. The retained masks provide the predictions evaluated below; the archived template history

is only partially documented. The control contains 48 two-hop, 30 three-hop and 22 four-hop questions.

The held-out question identifiers and exact resolved-subquestion–passage pairs are absent from the training set. Within the training set, questions sharing an exact repeated pair are assigned together to prevent that pair from appearing in both fitting and evaluation folds.

4.3.2 Textual and Contextual Features

For each candidate cell, a 193-dimensional feature vector describes the relation between the resolved subquestion and the supporting passage. The features combine passage-level and sentence-level ColBERT scores, lexical similarity, token alignment, title overlap and features of the subquestion before intermediate-answer substitution. ColBERT supplies contextual token representations with a late interaction score [2].

Absolute support and relative support are both represented. Background normalisation compares semantic scores against a fixed pool of 300 passages drawn from training data. Row features compare one subquestion’s support for different evidence passages. Column features compare different subquestions’ support for the same passage. Agreement features record whether different similarity measures identify the same candidate.

These comparisons are relevant to ownership: two passages may share an entity, while only one answers the relation requested by the subquestion. They also permit a cell to receive low support even when it is the best option in its row. The predictor can consequently leave a row or column empty. Retrieval rank, measured depth, reference hop position and the diagonal reference label are excluded from the feature vector. The predictor is fitted to textual assignment labels rather than to the retrieval outcomes that it will help evaluate.

The evaluation still supplies the annotated evidence passages as candidates. Thus the mask is an offline attribution instrument with known evidence to assign. Using text-based features does not by itself provide the complete evidence set for a new question at deployment.

4.3.3 Fitting, Threshold Selection and Ensemble Decision

Five grouped folds are constructed while balancing reference hop cohorts. Each rotation uses three folds to fit a weighted logistic model, one disjoint fold to select its threshold and one fold for evaluation. Regularisation is selected within the fitting portion. Standardisation parameters are estimated from the same fitting data. The retained configuration gives positive ownership examples weight 3.

For model b , let $p_b(a, i)$ be its predicted ownership score. Its threshold is obtained from the lower tail of scores assigned to positive calibration cells. If those scores in ascending order are $p_{(1)}, \dots, p_{(m)}$, the rule is

$$t_b = p_{(\lfloor (m+1)\alpha \rfloor)}, \quad \alpha = 0.01, \quad (4.10)$$

with threshold $-\infty$ when the indicated index is zero. The final mask uses a majority vote:

$$\widehat{M}_{ai} = \mathbf{1} \left\{ \sum_{b=1}^5 \mathbf{1}\{p_b(a, i) \geq t_b\} \geq 3 \right\}. \quad (4.11)$$

The lower-quantile threshold favours retaining true ownership assignments. Here α specifies the threshold-selection rule. Empirical recall, precision and complete-mask agreement quantify the resulting five-model procedure. Recall and precision are computed over mask cells pooled across questions, complete-mask agreement is the fraction of questions whose predicted mask matches the reference in every cell, and an arm row is exact when the prediction matches the reference in every cell of that row. A per-cell quantile rule and a majority vote require additional assumptions and analysis to yield a simultaneous guarantee for every assignment in a question. The results in this chapter are held-out measurements of the specified ensemble.

4.3.4 How Assignment Error Changes the Depth Measurement

Ownership prediction affects the measurement itself. Adding a positive cell adds a retrieval obligation and can increase the masked maximum. Removing a positive cell can reduce that maximum if the corresponding passage retains another owner. Removing its last owner instead censors the question. Mask errors can therefore move measured depth in either direction.

This dependence motivates evaluation at three levels: individual cells, complete arm rows and complete question masks. It also motivates freezing the attribution model when comparing decomposition prompts. A change in attribution must be distinguished from a change in the underlying retrieved lists.

4.4 Experimental Protocol

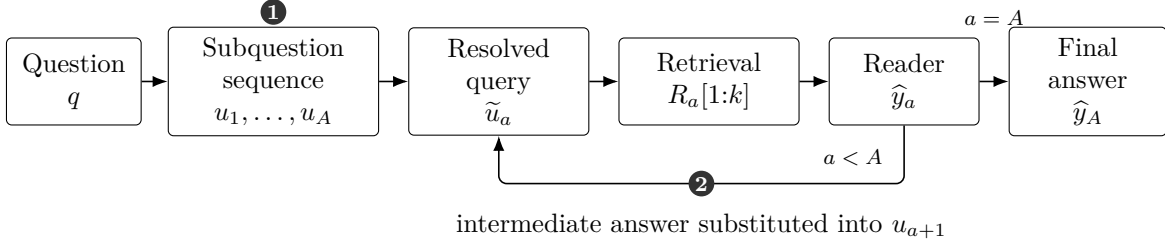
4.4.1 Systems and Retained Executions

The retrieval study uses the same 100 MuSiQue questions across the compared configurations. Its two language-model settings use GPT-4o-mini and Llama-3.3-70B-Instruct respectively for decomposition and reading. Retrieval uses HippoRAG 2 with NV-Embed-v2 embeddings and the model-specific graph and reranking configuration. HippoRAG 2 combines passage information with graph retrieval based on personalised PageRank [3]. Each reader call receives five passages; the retained rankings extend to 200 passages for depth analysis.

Three generated-decomposition prompts are compared with direct full-question retrieval. The baseline decomposition prompt serves as the reference generated policy, while the atomic-query prompt requests atomic single-fact queries. Both prompts use the same three demonstrations. The extended-instruction prompt uses a more detailed instruction specification without those demonstrations. These are different prompting configurations, so their comparison measures the full prompting choice rather than instruction length alone.

Two GPT-4o-mini controls use the reference decomposition. The first substitutes gold intermediate answers into downstream queries. The second substitutes the reader’s predicted intermediate answers. Both retain reference subquestions; the gold-answer condition isolates the effect of supplying correct intermediate bindings within this pipeline.

Figure 4.1 shows the execution shared by these configurations and states what each condition supplies at the two points that the comparison varies.



Condition	① Subquestion sequence	② Intermediate bindings
Direct	None: the full question is the only query	None: one retrieval and one reader call
Generated plan	Prompt-generated (baseline, atomic-query, extended-instruction)	Reader's predicted intermediate answers
Reference plan (predicted)	MuSiQue reference decomposition	Reader's predicted intermediate answers
Reference plan (gold)	MuSiQue reference decomposition	Gold intermediate answers (oracle-assisted)

Figure 4.1: Execution conditions in the retrieval study. The diagram is the execution shared by every condition, with the two marked points that the comparison varies: the source of the subquestion sequence and the source of the intermediate answers substituted into later subquestions. The table states what each condition supplies at those two points. Direct execution has neither, and the gold-binding condition is oracle-assisted.

4.4.2 Metrics and Comparison Units

Answer quality is measured by exact match (EM) and token F1 against the reference answer and its accepted aliases, under the benchmark’s answer normalisation: lowercasing, removal of articles and punctuation, and collapse of whitespace [1]. Let $t(\cdot)$ denote the token multiset of a normalised string, and let $\mathcal{A}(y)$ denote the reference answer y together with its accepted aliases. Then

$$\text{EM}(\hat{y}, y) = \mathbf{1}\{\hat{y} \in \mathcal{A}(y)\}, \quad \text{F1}(\hat{y}, y) = \max_{y' \in \mathcal{A}(y)} \frac{2|t(\hat{y}) \cap t(y')|}{|t(\hat{y})| + |t(y')|}, \quad (4.12)$$

where all compared strings are normalised and $\cap$ is the multiset intersection. Both quantities are averaged over the panel; EM is reported as a percentage and F1 on a 0–100 scale.

Retrieval is measured by complete-evidence coverage (Equation 4.8), union depth (Equation 4.3) and ownership-masked depth (Equation 4.4), summarised by the depth score J_D of Equation 4.9. The number of generated arms, retrieved passage slots and recorded total model tokens describe resource use. A passage slot is one passage delivered to one reader call. Per-arm passage counts therefore charge duplicate passages again when they are sent to separate reader calls, while unique passage counts describe the pooled collection. Recorded tokens are the prompt and completion tokens of the decomposition and reading calls of a question.

The reference-answer control records one token total across several execution branches. Per-branch costs are therefore unavailable. Generated and direct runs have separate recorded totals, which are reported per question.

Comparisons pair the same questions across configurations. Where intervals are reported, they are percentile intervals from 10 000 paired question-level bootstrap resamples.

They describe variation within this development panel. Several prompt and attribution configurations were examined on the panel, so these intervals are descriptive rather than a correction for adaptive model selection. The reference-style ownership control was kept separate from weight fitting and threshold calibration; its role is a structured assignment control rather than an evaluation of arbitrary generated plans.

4.5 Results

4.5.1 Ownership Recovery on Reference-Style Arms

Table 4.1 reports the retained ownership ensemble on the reference-style control. Of 274 reference owner cells, 269 are recovered. Seven non-owner cells are selected. Precision is 97.46% and recall is 98.18%. The ensemble recovers all assignments exactly on 92 of the 100 questions. It recovers 264 of the 274 complete arm rows, which is a different unit from complete question-mask agreement.

Table 4.1: Ownership prediction on the reference-style decomposition control. Cell metrics use 814 subquestion–passage pairs. Complete-mask agreement is measured over the 100 questions.

Measurement	Value
True positives / false negatives	269 / 5
False positives / true negatives	7 / 533
Precision	97.46%
Recall	98.18%
Cell accuracy	98.53%
AUROC of mean model scores	0.9977
Exact arm rows	264/274
Exact question masks	92/100

The cohort breakdown in Table 4.2 shows the distribution of the twelve cell errors. Four-hop questions contain three of the five missed owners. Because several cells belong to the same question, cell totals are reported with their question counts rather than interpreted as independent question outcomes.

Table 4.2: Ownership errors by reference hop count. TP, FN, FP and TN count individual cells, while n counts questions.

Reference hops	n	Cells	TP	FN	FP	TN
2	48	192	95	1	4	92
3	30	270	89	1	0	180
4	22	352	85	3	3	261

On the gold-answer retrieval traces, the exact reference mask yields complete assigned-evidence coverage of 100% at depth 200 and a score of 86.47. Using the learned ownership mask gives 97% coverage and a score of 84.84. This comparison isolates the effect of replacing the reference assignment with its prediction on those fixed lists.

4.5.2 Sensitivity to the Attribution Model

Table 4.3 scores the same retrieval traces under two attribution models: an earlier predictor built on a different feature set, and the retained model of Section 4.3. The baseline generated decomposition receives scores of 52.32 and 72.19 respectively. The analogous atomic-query values are 52.81 and 66.64. These changes arise from the assigned cells, with the passages and ranks held fixed. On the reference arms with gold bindings the two models differ by 24.98 points, while replacing the retained prediction with the exact reference assignment changes the score by 1.63. The masked-depth measurement is therefore more sensitive to the choice of attribution model than to the residual error of the retained one.

Table 4.3: Masked-depth measurements under two attribution models on fixed GPT-4o-mini retrieval traces. Coverage is complete assigned-evidence coverage at $K = 200$; score is J_{D_M} from Equation 4.9. The passages and their ranks are identical across rows, so only the assigned cells differ. The last row uses the exact reference assignment.

Decomposition	Attribution	Coverage (%)	Score
Baseline	Earlier model	78	52.32
Baseline	Retained model	87	72.19
Atomic-query	Earlier model	77	52.81
Atomic-query	Retained model	81	66.64
Reference, gold bindings	Earlier model	88	59.86
Reference, gold bindings	Retained model	97	84.84
Reference, gold bindings	Reference assignment	100	86.47

The reference-style control supports the use of the retained predictor for recovering known assignments. For generated arms, assignments can legitimately be fused or shared. The score differences in Table 4.3 therefore establish sensitivity to attribution rather than a labelled accuracy comparison on generated plans. Prompt comparisons in Chapter 5 keep the current ownership model fixed.

4.5.3 Evidence Availability and Final Answers

For readability, $C_U(k)$ denotes the union coverage $C_{D_{\text{union}}}(k)$ in the result tables. Table 4.4 reports the direct and generated-decomposition executions. With GPT-4o-mini, the baseline decomposition raises complete union coverage at five passages per arm from 45% to 76%. EM changes from 40% to 52% and F1 from 52.64 to 59.00. With Llama-3.3-70B, complete union coverage changes from 46% to 79%, while EM changes from 42% to 43% and F1 from 52.30 to 51.37.

The paired bootstrap interval for the baseline decomposition’s EM change is $[2, 22]$ percentage points for GPT-4o-mini and $[-8, 10]$ for Llama-3.3-70B. These measurements show why retrieval and answer metrics are both needed. Improved pooled availability can coexist with a small change in final answers, because the sequence must still use the retrieved evidence and propagate useful intermediate states.

The per-arm comparison also changes the total retrieval expenditure. The baseline GPT-4o-mini decomposition retrieves a mean of 13.95 passage slots across its reader calls, compared with 5 for direct retrieval. Its mean pooled collection contains 11.35 distinct passages. Thus the coverage increase in Table 4.4 combines a change in query

Table 4.4: Recorded execution results on 100 MuSiQue questions. EM and F1 are percentages. $C_U(5)$ is complete union coverage at five passages per arm. Arms and tokens are per-question means. Retrieved evidence and final answers are separate outcomes.

Model	Prompt	EM	F1	$C_U(5)$	Arms	Tokens
GPT-4o-mini	Direct	40	52.64	45	1.00	4 369
	Baseline	52	59.00	76	2.79	12 763
	Atomic-query	52	59.66	72	2.66	11 898
	Extended-instruction	50	60.39	74	2.61	13 881
Llama-3.3-70B	Direct	42	52.30	46	1.00	4 530
	Baseline	43	51.37	79	3.00	14 069
	Atomic-query	45	53.54	78	3.22	14 789
	Extended-instruction	47	55.94	75	2.84	15 271

formulation with more passage slots and more reader calls. Section 4.6 compares fixed lists at a common total passage budget.

4.5.4 The Role of Intermediate Bindings

The two reference-decomposition controls use the same subquestion templates and reader but change the answers substituted into later queries. Table 4.5 reports their outcomes. Gold substitution yields EM 64% versus 49% under predicted substitution, with a paired bootstrap interval of $[8, 22]$ percentage points for the difference. Complete union coverage at five passages per arm changes from 80% to 92%.

Table 4.5: Reference-decomposition controls with GPT-4o-mini. Gold bindings supply reference intermediate answers when constructing downstream queries. Predicted bindings use the reader’s earlier outputs.

Intermediate substitution	EM (%)	F1	$C_U(5)$ (%)
Predicted answers	49	57.50	80
Gold answers	64	70.73	92

The intervention changes downstream query text and consequently can change both retrieval and answer generation. Its effect includes error propagation through those two stages. Supplying gold intermediate answers defines an oracle-assisted control. The measured gap identifies a role for intermediate state quality within the reference-chain pipeline; it is not a gain achieved by an operational binding policy.

4.6 Depth Under a Total Passage Budget

4.6.1 Equal-Allocation Replay

A common per-arm depth gives more total passage slots to longer decompositions. To separate these quantities, consider a per-question ceiling $B \leq AK$ distributed among the A recorded retrieval lists. Equal allocation uses the whole allowance and assigns

$$k_a(B) = \left\lfloor \frac{B}{A} \right\rfloor + \mathbf{1}\{a \leq B \bmod A\}, \quad \sum_{a=1}^A k_a(B) = B. \quad (4.13)$$

The remainder is allocated in execution order, a tie-break over at most $A - 1$ slots that favours the earlier arms whose answers resolve later queries. A passage slot is charged each time it occurs in an arm prefix, including duplicates across arms. Complete evidence availability is then evaluated in $\bigcup_a R_a[1 : k_a(B)]$.

This allowance counts selected passage slots. Equal slot counts can produce different numbers of unique passages and different token totals, since passages can recur across arms and vary in length. Retrieving the ranked lists, producing the queries and reading the contexts incur separate computational costs.

These are fixed-list replays. The original intermediate answers and later queries were generated using five passages per reader call. Changing a prefix in the replay leaves those earlier answers and later queries unchanged. The result measures the coverage of the recorded lists at a specified slot budget. An end-to-end policy evaluated at that budget would regenerate intermediate answers and dependent queries under its actual contexts.

Per-question and average budgets. Under a hard ceiling B for each question, the largest feasible common depth within a question is $\lfloor B/A(q) \rfloor$. Complete-evidence availability is non-decreasing in every prefix length, so this choice weakly dominates smaller equal depths. Equation 4.13 also spends the integer remainder. A single depth shared across the entire panel is a weaker comparator when arm counts vary: feasibility limits it to $\lfloor B/\max_q A(q) \rfloor$. Unequal within-question allocation can improve on the balanced comparator by placing more slots on deeper arms. An average budget instead permits transfers between questions, including transfers that retain positive allocations for every question. That is a different resource constraint from the per-question comparisons here.

4.6.2 A Fixed-List Oracle Allocation

The rank matrix also permits an exact comparison with the least expensive prefix allocation for its recorded lists:

$$B^*(q) = \min_{k_1, \dots, k_A \in \{0, \dots, K\}} \left\{ \sum_a k_a : E(q) \subseteq \bigcup_a R_a[1 : k_a] \right\}. \quad (4.14)$$

The value is $+\infty$ when the retained lists jointly miss annotated evidence. For finite cases, it can be computed by dynamic programming over subsets of evidence. Let $S_a(k)$ denote the evidence indices found in arm a 's first k passages. Starting from $F_0(\emptyset) = 0$, the update is

$$F_a(T \cup S_a(k)) \leftarrow \min\{F_a(T \cup S_a(k)), F_{a-1}(T) + k\}. \quad (4.15)$$

Unreachable states have value $+\infty$, and $B^*(q) = F_A(\{1, \dots, h\})$. Only zero and prefix lengths at which an additional evidence item appears need to be considered: a longer prefix with the same evidence set adds cost without changing that state. With at most four annotated hops in this panel, the subset state space is small.

The oracle uses gold evidence identities to choose its allocation. Moreover, all recorded queries are available even when their selected prefix length is zero. The objective charges passage slots but leaves the generation of those queries and their intermediate states outside the optimisation. It is an exact minimum for the fixed-list problem and an optimistic reference for a system that must construct and execute the queries. It is not an achieved runtime or token saving.

4.6.3 Allocation Headroom on the Recorded Lists

The budget $B = 15$ is close to the mean of 13.95 passage slots recorded for the baseline GPT-4o-mini decomposition, so the replay compares configurations at approximately the cost of that decomposition. At $B = 15$, equal allocation recovers complete evidence for 76 questions under the baseline GPT-4o-mini decomposition and 78 under its Llama counterpart. The corresponding oracle allocations reach 85 and 87; an oracle entry counts the questions whose minimum feasible budget $B^*(q)$ in Equation 4.14 is at most 15. Table 4.6 reports the same comparison for the other prompts and for direct retrieval, which are the configurations available without a reference decomposition. Direct retrieval has a single arm, so its two allocations coincide by construction, and its entries give that one list 15 passages rather than the five used in execution.

Table 4.6: Complete evidence coverage (%) at a total budget of 15 passage slots on fixed recorded lists. The oracle chooses prefixes using gold evidence and is a reference allocation rather than a deployed policy. Each row contains 100 questions.

Model	Prompt	Equal allocation	Oracle allocation
GPT-4o-mini	Direct	66	66
	Baseline	76	85
	Atomic-query	73	85
	Extended-instruction	75	83
Llama-3.3-70B	Direct	65	65
	Baseline	78	87
	Atomic-query	75	88
	Extended-instruction	75	85

The baseline decompositions retain a coverage advantage over direct retrieval in this replay after the total slot budget is matched. The additional nine questions recoverable by oracle allocation on the baseline decomposition in each model setting identify uneven depth requirements across arms. This result motivates evaluation of observable allocation policies. Their value must be measured after accounting for the decisions and intermediate answers needed to obtain the lists.

4.7 A Cross-Benchmark Fixed-Trace Study

4.7.1 Retrieval Regime and Evaluation Population

A complementary BM25 study applies the passage-slot accounting to MuSiQue, 2Wiki-MultiHopQA [5] and HotpotQA [4]. It uses separate closed corpora containing 11 656, 10 426 and 9 811 passages, respectively. Supporting passages are identified from each benchmark’s annotations. The metric is complete annotated-evidence availability in the selected union; it requires the evidence set but no estimated ownership mask. The ownership measurement remains relevant when the question is which arm has the evidence needed for its own decision.

Each source panel contains 1 000 questions. Saved generated-plan traces are available for 824 MuSiQue, 977 2Wiki and 996 HotpotQA questions. The full-question control is restricted to the same retained IDs within each dataset. Thus the comparisons are paired on available traces, with 176, 23 and 4 source questions outside their respective

denominators. These coverage values are conditional on obtaining a retained trace; they do not represent success rates for the entire original intake. The source panels also have historical project exposure, so these results are development evidence separate from the frozen comparison in Chapter 5.

The sequential traces use generated subquestions. A GPT-4o-mini reader answers each arm from five BM25 passages, and those predicted answers resolve later references. Up to 200 ranks per arm are saved for replay. This is a different retriever and execution implementation from the HippoRAG experiments above. Every allocation below acts on those saved lists. Its slot budget excludes the calls and contexts used to generate the intermediate answers.

Selected slots, unique pooled passages and reader tokens remain distinct. Two allocations with the same $\sum_a k_a$ can have different overlap between arms and hence different deduplicated context sizes. A subsequent reader over the pooled union is an additional policy; its answer quality must be measured separately from the coverage frontiers reported here.

4.7.2 Matched Passage-Slot Frontiers

Table 4.7 compares full-question retrieval, balanced allocation from Equation 4.13, and the fixed-list oracle at $B = 8$. The balanced allocation spends the integer remainder in execution order. The saved plans contain at most seven arms on MuSiQue and HotpotQA and six on 2Wiki, so this budget can allocate at least one slot to every arm.

Table 4.7: Complete annotated-evidence coverage at a ceiling of eight selected passage slots. BM25 results use fixed predicted-binding traces; the oracle uses gold evidence ranks. Counts are retained questions from each original 1,000-question panel.

Dataset	n	Full question	Balanced	Oracle
MuSiQue	824	0.136	0.413	0.536
2Wiki	977	0.315	0.688	0.881
HotpotQA	996	0.637	0.762	0.836

The oracle recovers 12.4, 19.3 and 7.4 additional percentage points over balanced allocation on MuSiQue, 2Wiki and HotpotQA, respectively. These are opportunities within the existing rankings. They preserve the recorded queries and reveal how unequal evidence ranks constrain a balanced split. The gain from generating better queries is outside this oracle’s policy class.

The subset dynamic program in Equation 4.15 independently reproduces the saved oracle coverage on all three retained panels at the checked budgets. For a fixed panel, its frontier is

$$C^*(B) = \frac{1}{n} \sum_{q=1}^n \mathbf{1}\{B^*(q) \leq B\}. \quad (4.16)$$

Questions whose lists jointly miss a required passage remain uncovered for every allocation. Zero-depth arms are permitted by the oracle, with the cost of constructing their saved queries kept outside the objective.

A complementary reading fixes coverage and compares the first tested budget that reaches it. On the MuSiQue sequential panel, balanced allocation first reaches coverage 0.50, 0.60 and 0.70 at 16, 32 and 128 slots. The oracle reaches the same targets at 8, 16 and 48 slots. These values are minima over the reported budget grid, rather than exact

interpolated crossings. They quantify allocation headroom and do not constitute achieved token or runtime reductions.

Generated arm count also differs from benchmark reasoning depth. Mean arm counts are 2.53, 2.79 and 2.10 on the retained MuSiQue, 2Wiki and HotpotQA panels. More arms create more allocation choices and can increase the cost of a common per-arm depth. They can also reflect redundant splitting or a prompt preference. An association between allocation headroom and arm count therefore concerns plan structure, rather than establishing that semantically harder questions obtain greater savings.

4.7.3 Reference and Generated Query Regimes

The MuSiQue artifacts support three retrieval regimes. Reference plans use the benchmark decomposition and gold intermediate answers. Sequential plans use generated subquestions and the reader’s predicted answers. Paraphrased queries replace a reference with a noun phrase derived from the referenced subquestion, without reading an intermediate answer. Table 4.8 reports their balanced coverage at two total slot budgets.

The sequential and paraphrase archives contain the same 824 question IDs and the same arm counts. Their different rankings show that the way intermediate references are turned into query text materially changes evidence availability. Reference plans provide a separate oracle-assisted comparison: they change both the source of the plan and its bindings, and their full reported panel has a different size. The three-row comparison therefore does not isolate a pure binding intervention. The controlled reference-chain comparison earlier in this chapter supplies the narrower evidence about predicted versus gold intermediate answers.

The distinction also constrains interpretation of allocation. At $B = 200$, the sequential fixed-list oracle covers 0.795 of the retained MuSiQue questions, compared with balanced coverage 0.733. Both results concern the selected prefixes of queries already produced with five passages per intermediate reader. A changed operational budget can alter those answers and later rankings, which requires re-execution to evaluate.

4.8 Discussion and Implications for Optimisation

The evaluation decomposes a retrieval plan into measurable obligations. Union depth measures pooled evidence availability. The ownership mask localises that evidence to the subquestion expected to use it. Answer metrics then measure the outcome of executing the sequence. These distinctions prevent a shallow union from being interpreted as a fully supported reasoning chain and prevent a masked-depth change caused by attribution from being attributed to improved retrieval.

Table 4.8: MuSiQue BM25 retrieval regimes. Coverage is measured on each retained panel under balanced slot allocation. The reference row uses a different plan source and a larger panel.

Query regime	n	Coverage at $B = 8$	Coverage at $B = 200$
Reference plan, gold bindings	1 000	0.595	0.962
Generated plan, predicted bindings	824	0.413	0.733
Generated plan, paraphrased references	824	0.109	0.397

The ownership study provides a working measurement model. It recovers the reference assignment on 92 of 100 complete reference-style masks while retaining high cell recall and precision. Its empirical scope is reference-style arms with known supporting passages. Generated plans can change the semantic structure through fusion, splitting, omission or incorrect dependency substitution. Chapter 5 evaluates controlled transformations of reference plans to measure sensitivity to several of these changes. Transfer to naturally generated plans requires independently labelled assignments from that regime.

The answer experiments establish the importance of execution state. Under reference decompositions, substituting gold intermediate answers changes both retrieval coverage and final accuracy. An unowned passage in the predicted mask can therefore have several explanations: a missing information need, an altered query following an earlier error, or a classifier disagreement. Feedback for optimisation should retain those distinctions. A predicted attribution diagnosis is evidence to inspect, rather than a verified causal explanation of the failure.

The resource measurements distinguish retrieval depth, total passage slots and model tokens. The masked objective rewards shallow completion of assigned evidence; the fixed-budget replay measures allocation efficiency on existing lists. A prompt that raises the former can still create more arms or consume more tokens. Chapter 5 consequently evaluates the optimisation objective together with complete-evidence coverage, answer quality and execution cost.

The complementary BM25 study extends the fixed-slot analysis to three benchmarks and larger retained panels. Its oracle gaps motivate observable allocation policies, while its query-regime comparison establishes the need to document how dependent subquestions were resolved. Those results retain their closed-corpus, fixed-trace and available-case scope alongside the separate attribution and answer measurements.

The current system applies decomposition to every question in these experiments. The difficulty probe of Chapter 3 supplies a separate measurement and is not used to select or route these executions. Connecting a pre-answer difficulty estimate to a decomposition policy requires its own cost and answer evaluation.

Conclusion

This chapter addressed the evaluation component of the retrieval-efficiency question posed in Section 2.6.2. It defined complete evidence availability and ownership-masked retrieval depth, trained a textual ownership predictor and evaluated its assignments on reference-style decompositions. The predictor recovered 269 of 274 owner cells and 92 complete question masks out of 100.

The retrieval study showed that evidence availability, assignment and answer execution vary separately. Generated decompositions increased complete pooled retrieval coverage, while the answer gains depended on the model and execution state. A fixed-budget analysis also identified room to allocate passage slots more selectively across the recorded arms. These findings support an evidence-aware optimisation objective while specifying the additional answer and cost measurements required to assess its practical effect. Chapter 5 uses this evaluation to study decomposition prompt search and learned passage allocation.

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